2025 Oregon Mechanical Specialty Code

Code review committee matrix

This matrix is a summary of the code review committee's recommendation to the Mechanical Board for the adoption of the 2025 OMSC.

Abbreviations:

ORA Existing/New Oregon amendment
IMC International Mechanical Code change
IFGC International Fuel Gas Code change
SAM Statewide Alternate Method
COMM Committee discussion
***ICC** Fiscal impact reported by the original ICC proponent

Section	Topic / Summary of changes	Outcome	Fiscal
CHAPTER 2	**DEFINITIONS**		
All	**All defined terms.**		
	ORA Aligned with existing Oregon amendments.	**Retain ORA**	—
CHAPTER 3	**GENERAL REGULATION**		
306.5	**ORA** Existing amendments: Clarifies item 10 that "access paths to ladders and ladder landings shall remain unobstructed at all times." Adds an exception for the replacement, repair or maintenance of an existing appliance or piece of equipment lawfully.	**Retain ORA**	—
	IMC M13-21, M14-21: Alignment with 2018 OSHA. **ICC: Potential cost increases as some of the dimensions have increased such as side rail height etc.*	**Approve IMC**	*No fiscal
307.2.1	**Condensate disposal.**		
	ORA Existing amendments: Adds three possible approved places of disposal.	**Retain ORA**	—
307.4	**Condensate drain subject to freezing.**		
	ORA New amendment: Add a section for condensate drains from cooling/evaporator coils that operate year-round and can produce condensate in winter (subject to freezing).	**Approve New ORA**	*No fiscal
CHAPTER 4	**VENTILATION**		
401.2	**Ventilation required.**		
	ORA Rescind: OMSC pointed to OSSC for natural ventilation. Now, the OSSC will reference the OMSC for requirements.	**Rescind OR A**	
	Rescind: The OMSC previously removed the sentence regarding dwelling units.	**Rescind ORA**	
	New amendment: Adopt ventilation in dwelling unit requirement, but remove reference to Energy Code and ASHRAE 90.1, it is not necessary.	**Approve New ORA**	*No fiscal
	Existing amendment: Note pointing to OEESC for heating and air-conditioning controls.	**Retain ORA**	
402	**Natural ventilation.**		
	ORA Rescind: The OMSC pointed to the OSSC for natural ventilation in the past. Now, the OSSC will reference the OMSC for requirements, so the previous note not adopting Section 402 is rescinded. IMC Sections 402.1 – 402.4 are adopted.	**Rescind ORA** **Adopt IMC**	*No fiscal
403	**Mechanical ventilation.**		
	ORA New and existing amendments: • 403.2.2 Transfer air: Retaining existing amendments • 403.3 Outdoor air and local exhaust airflow rates: IMC removed reference to "three stories and less in height above grade-plane and sends all Group R-2, R-3 and R-4 occupancies to 403.3.2. • Table 403.3.1.1 Minimum Ventilation Rates Existing ORA - FN (g) replaces "For occupancies other than science labs" and adding "for environmental air exhaust." Adds FN (n) and (m) for toilet rooms and bathrooms in private dwellings, single and multiple. M20-21, M21-21, M22-21: Added adult changing station. Updated ventilation rate to ASHRAE Standard 62.1-2019. • 403.3.2.1 Outdoor air for dwellings units: M19-21 Revised the equation from including $0.01A_{floor}$ to $0.03A_{floor}$ BCD recommends changing it to $0.02A_{floor}$ • Table 403.3.2.3 Minimum Required Local Exhaust Rates for Group R-2, R-3 and R-4 occupancies: BCD recommends adding the FN from Table 403.3.1.1 regarding All rooms containing bathing or residential spa facilities with intermittent exhaust shall be provided with a mechanical ventilation system controlled by a dehumidistat, timer or similar means of automatic control.	**Approve IMC as modified by ORA**	*Cost increase $0 - $400 for 0.02 $0 - $800 for 0.03.
407.1	**Ambulatory Care facilities and Group I-2 Occupancies.**		
	ORA Existing amendments: Oregon modifies to align with OAR 333 and licensure requirements of OHA.	**Retain ORA**	—

Section	Topic / Summary of changes		Outcome	Fiscal
404.1	**Enclosed parking garages.**			
	ORA	Existing amendments: Oregon references the Energy Code and deleted item no. 1.	**Retain ORA**	—
CHAPTER 5	**EXHAUST SYSTEMS**			
501.3.1	**Location of exhaust outlets.**			
	IMC	M16-21, M28-21: Factory-built intake/exhaust combination termination fittings are regularly provided by manufacturers and installed by builders to separate mechanical air intakes from mechanical exhaust serving dwelling unit or sleeping unit mechanical ventilation systems. Special approval should not be required for these types of fittings when installed in accordance with the manufacturer installation instructions.	**Approve as modified**	*No fiscal
	COMM	*Committee recommends approving the IMC change as modified by adding an option to use ASHRAE 62.1 Normative Appendix B.*		
501.6	**Common ducts.**			
	IMC	M29-21 Part I: Added provisions addressing common ducts serving separate dwelling or sleeping units. Exhaust ducts that are under positive pressure cannot be joined because the airflow from one fan will leak out through the fan that is not running, so they shall be connected where maintained at negative pressure. **ICC: Potential code increase only where the cost of an additional roof, wall or soffit penetration is more than the cost of larger common ducts, tee and wye fittings, fasteners, sealants and hangers and the extra labor to assemble common duct arrangements. Potential code increase if additional roof or wall penetrations cost more than combined exhaust discharge ducts. Combining ducts into a common duct adds material costs, as does making roof and wall penetrations.*	**Approve IMC**	*Cost increase
504.9.2	**Duct installation.**			
	ORA	Existing amendments: Removes the last sentence regarding screws or similar fasteners in domestic duct installation.	**Retain ORA**	—
504.9.3	**Transition ducts.**			
	ORA	Existing amendments: Removes UL 2158A reference.	**Retain ORA**	—
505.2	**Exhaust ducts for domestic cooking exhaust equipment.**			
	ORA	Existing amendments: Removes exception for natural ventilation.	**Retain ORA**	—
505.6	**Other than Group R.**			
	ORA	Existing amendments: Deletes section to align with 507.1.2.	**Retain ORA**	—
505.7	**Group I-1 occupancies.**			
	IMC	M32-21: New section addressing hood installations over domestic cooking equipment in Group I-1 occupancies. **Note: The code currently allows for this but does not require 500 cfm. Potential cost increase for the 500-cfm requirement.*	**Approve IMC**	*Cost increase
505.8	**Group I-2 occupancies.**			
	IMC	M32-21: New section addressing hood installations over domestic cooking equipment in Group I-2 occupancies. **Note: The code currently allows for this but does not require 500 cfm. Potential cost increase for the 500-cfm requirement.*	**Approve IMC**	*Cost increase
506.3.2.5	**Grease duct test.**			
	IMC	M38-21: Added two new sections with light-test and water-spray test provisions.	**Approve IMC**	*No fiscal
	ORA	Existing amendments: Adds approved equivalent test as an alternative to a light test.	**Retain ORA as modified**	*No fiscal
	COMM	*Retain the ORA as modified by removing "pressure" and adding the "water spray test" as an option.*		
507.1	**Commercial Kitchen Hoods.**			
	IMC	M40-21: Reorganized Section 507 into a logical order of application, while not making any substantive technical changes. M41-21: Revised to allow the Type I hood if it is installed in full compliance with Type I hood installation requirements. M42-21: Revised to clarify the three existing exceptions for Section 507.1 where because of the particular type of product or cooking operation, a hood is not required above the product or cooking operation. M43-21: Added to the exceptions: "Ovens listed and labeled for use with wood fuel in accordance with UL 2162 and vented in accordance with the manufacturer's instructions." M44-21: The exception from 507.2 for electric cooking appliances was moved to the exceptions in Section 507.1 and the UL standard was changed to UL 197. M45-21: Added to the exceptions: "Commercial electric dishwashers incorporating a self-contained condensing system *listed* and *labeled* in accordance with UL 921." M46-21: Where the heat and moisture loads from dishwashers and appliances that produce heat or moisture and do not produce grease or smoke as a result of the cooking process are incorporated into the HVAC system, a Type II hood above is not needed.	**Approve IMC**	*No fiscal
	ORA	New amendment: Revises the new provisions regarding space containing those cooking spaces that do not require Type II hoods to comply with the Oregon-specific exception in Section 507.3.	**Approve New ORA**	*No fiscal

Section	Topic / Summary of changes		Outcome	Fiscal
507.1.2	**Domestic cooking appliances used for commercial purposes.**			
	ORA	Existing amendments: Adds an exception for alignment.	**Retain ORA**	—
CHAPTER 6	**DUCT SYSTEMS**			
601.5	**Return air openings.**			
	IMC	M51-21: Revised to clarify the intent of this section without changing its meaning. Section 601.5 establishes requirements for return air for "heating, ventilation, and air-conditioning systems" that shall not be taken from a closet, bathroom, toilet room, kitchen, garage, boiler room, furnace room or unconditioned attic. M53-21: Added further requirements for return air from a closet to mitigate mold growth in closets.	**Approve IMC**	*No fiscal
T. 603.4	**Duct construction minimum sheet metal thickness for single dwelling units.**			
	ORA	Existing amendments: Oregon retains the 2014 OMSC table.	**Retain ORA**	—
CHAPTER 7	**COMBUSTION AIR**			
No changes were made to Chapter 7.			**N/A**	—
CHAPTER 8	**CHIMNEYS AND VENTS**			
No changes were made to Chapter 8.			**N/A**	—
CHAPTER 9	**SPECIFIC APPLIANCES, FIREPLACES AND SOLID FUEL-BURNING EQUIPMENT**			
907.1	**Incinerators and crematories.**			
	IMC	FG7-21 Part II: Added listing and label requirements for factory-built cremation furnaces and commercial direct-fed incinerators with UL 2790. Clarified listing and labeling requirements for factory-built incinerators for domestic applications shall be listed and labeled in accordance with UL 791. Clarified installation requirements for incinerators and cremation furnaces in accordance with MII's. **ICC: No fiscal impact.*	**Approve**	*No fiscal
931	**Incinerators and crematories.**			
	IMC **ORA**	M62-21: Introduces requirements for steam bath equipment listings, and installation criteria. New amendment: Delete section/not under Mechanical Code scope.	**Disapprove IMC**	—
CHAPTER 10	**BOILERS, WATER HEATERS AND PRESSURE VESSELS**			
Chapter 10	**All of the Chapter.**			
	ORA	Existing amendments: Aligns with Plumbing code and Boiler code.	**Retain ORA**	—
CHAPTER 11	**REFRIGERATION**			
1101	**Technician certification requirements.**			
	ORA	Add note: Persons engaging in service, repair, or disposal of equipment that could release refrigerants into the atmosphere shall be certified under Environmental Protection Agency Section 608 Technician Certification requirements under 40 CFR Part 82, Subpart F. Section 608 Technician Certification Requirements | US EPA / Technology Transitions | US EPA	**Approve New ORA**	*No fiscal
1101.1.1	**Refrigerants other than ammonia.**			
	IMC	M69-21: Added "Refrigeration systems containing carbon dioxide as the refrigerant shall also comply with IIAR CO^2."	**Approve IMC**	*No fiscal
1101.7	**Changing refrigerant.**			
	OR A	**Existing amendments:** Deletes the section regarding changing refrigerant.	**Retain ORA**	—
1103.1	**Refrigeration standards**			
	SAM	21-01: Intended to allow use of newer, more environmentally friendly refrigeration methods using the most current refrigerant tech.	**Retain SAM**	—
1104.3.1, 1104.3.2	**Air conditioning for human comfort and Group A2, A3, B2 and B3 refrigerants.**			
	IMC	M75-21: Added changes in alignment with updated ASHRAE 15. **ICC: No fiscal impact.*	**Approve IMC**	*No fiscal
1105.9	**Machinery room means of egress.**			
	ORA	Existing amendments: Modify by pointing to the Building Code for provisions regarding machinery rooms larger than 1000 square feet, rather than duplicating the requirements in the OMSC.	**Retain ORA**	—
1108.1.1	**Installer certification.**			
	ORA	New amendments: Refrigerant piping joints and connections regulated under this code shall be made by persons with valid certification under ASME Section IX or AWS B2.2 for welding or brazing. Installer certification shall be presented to *building official* upon request.	**Approve New ORA**	*No fiscal

Section	Topic / Summary of changes	Outcome	Fiscal
1109.2.5	**Refrigerant pipe shafts.**		
	OR A New amendments: Deletes “using Group AI refrigerant” from the 2nd exception.	**Approve New ORA**	*No fiscal
CHAPTER 12	**HYDRONIC PIPING**		
1201.1	**Scope – Hydronic Piping.**		
	IMC M85-21: Adds radiant heating, radiant cooling, and snow- and ice-melting.	**Approve IMC**	*No fiscal
1209.1	**Materials.**		
	ORA Existing amendments: Oregon modified with added language “, cross-linked polyethylene (PEX) tubing, cross-linked polyethylene/aluminum/ cross-linked polyethylene (PEX-AL-PEX) pressure pipe or polybutylene rated.”	**Retain ORA**	—
1209.6	**Radiant tubing placement.**		
	IMC M97-21: Added sections addressing radiant tubing replacement.	**Approve IMC**	*No fiscal
1209.7	**Snow- and ice-melt tubing placement.**		
	IMC M98-21: Added sections addressing snow- and ice-melt tubing replacement.	**Approve IMC**	*No fiscal
CHAPTER 13	**FUEL OIL PIPING AND STORAGE**		
No changes were made to Chapter 13.		**N/A**	—
CHAPTER 14	**SOLAR THERMAL SYSTEMS**		
No changes were made to Chapter 14.		**N/A**	—
APPENDICES			
App-A	**Chimney connector pass-throughs.**	**No changes**	—
App-B	**Recommended permit fee schedule.**	**No changes**	—
App-C	**Board of appeals.** Oregon does not adopt.	**No adoption**	—
App-D	**Clean Air Delivery.** New IMC appendix. Oregon will not adopt.	**No adoption**	—
App-E	**Clean Air Delivery and Monitoring.** New IMC appendix. Oregon will not adopt.	**No adoption**	—
APPENDIX C	**FUEL GAS – OR A**		
C307.2	**Concealed Condensate Piping**		
	ORA Existing amendments: Adding that the secondary drain be clearly marked with an embossed or engraved tag or in an approved manner.	**Retain ORA**	—
C310	**Corrugated Stainless Steel Tubing**		
	ORA Existing amendments: Adds Section 310.1.1 Corrugated Stainless-Steel Tubing (CSST). CSST shall be bonded in accordance with the Electrical Code and manufacturer’s installation instructions.	**Retain ORA**	—
C404.7	**Protection of Piping against physical damage.**		
	ORA Existing amendments: Adds language to clarify that the exempted steel pipe must be at least schedule 40 or thicker.	**Retain ORA**	—